

In 1 dim $L: \mathbb{R} \rightarrow \mathbb{R}$ fixing a basis

$$\begin{array}{c} \nearrow \quad \nwarrow \quad \nearrow \\ \begin{array}{ccc} [a] & [x] & = [b] \\ \uparrow & \uparrow & \uparrow \\ \text{1x1 matrix} & \text{column length 1} & \end{array} \end{array} \quad L \text{ is invertible iff } a \neq 0$$
$$L^{-1} = \begin{bmatrix} \frac{1}{a} \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{a} \end{bmatrix} [a] [x] = \begin{bmatrix} \frac{1}{a} \end{bmatrix} [b] \Rightarrow [x] = \begin{bmatrix} \frac{b}{a} \end{bmatrix}$$

this $ax=b \quad x=\frac{b}{a}$ if $a \neq 0$

in 2 dim $L: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$\underbrace{\begin{bmatrix} a & b \\ c & d \end{bmatrix}}_L \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\text{column length 2}} = \underbrace{\begin{bmatrix} k \\ l \end{bmatrix}}_{\text{column length 2}} \quad L \text{ is invertible iff } ad-bc \neq 0$$

i.e. it has a non zero determinant

$L \quad \det(L) = ad-bc$

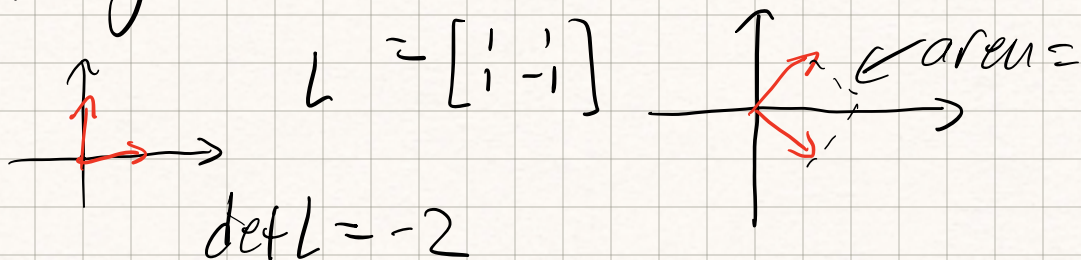
the inverse is $\frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

$$\frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ad-bc & db-db \\ -ac+ac & -bc+ad \end{bmatrix} \frac{1}{ad-bc} = I$$

The determinant tells us if a square matrix is invertible

Why would anyone think to look at the determinant?

in 2 dim

$$L = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$


$\det L = -2$

det is really the ^{signed} "area" of our n dimensional parallelepiped - orientation determines the sign of det

if $\hat{v} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ $\det = 0$  area = 0

$$\det \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = -2 \quad \det \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} = 2$$

Swapping rows (or columns) multiplies det by -1

$$\begin{vmatrix} 6 & 12 \\ 3 & 4 \end{vmatrix} = -12 \quad \begin{array}{l} \text{multiply row by const } K \\ \text{multiplies det by } K \end{array}$$

$$\begin{vmatrix} 1 & 2 \\ 1 & 0 \end{vmatrix} = -2 \quad \text{So adding multiples of a row to another doesn't change the det}$$

$$\begin{vmatrix} 2 & -1 & 6 \\ 4 & -2 & 3 \\ 1 & 1 & 1 \end{vmatrix} \xrightarrow{C_1 + 2C_2} \begin{vmatrix} 0 & -1 & 6 \\ 0 & -2 & 3 \\ 3 & 1 & 0 \end{vmatrix} = 6(6) - 9 = 27$$

$$\begin{array}{ccc|ccc} 2 & -1 & 6 & 1 & 0 & 0 \\ 4 & -2 & 3 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array} \xrightarrow{r_2 \leftrightarrow r_3} \begin{array}{ccc|ccc} 2 & -1 & 6 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \\ 4 & -2 & 3 & 0 & 1 & 0 \end{array}$$